

Hydrologic Response Units in OpenHydroQual

The HRU and Urban HRU composite blocks

Abstract

A hydrologic response unit (HRU) in OpenHydroQual is a composite block that bundles the surface, soil and groundwater stores of one piece of land and the connectors between them, so that a sub-catchment is added to a model as a single object with a single set of properties. This document describes the two HRUs: the `Hydrologic_Response_Unit` (“HRU”), a single infiltrating surface over a three-layer soil column and a groundwater cell, and the `Urban_HRU`, which divides the land into pervious and impervious surfaces with their own routing reaches, places the groundwater cell under the whole unit, recharges it through the unsaturated conductivity of the bottom soil layer, treats it as an unconfined aquifer, takes soil parameters layer by layer and carries two calibration multipliers. For each unit the document sets out the members and connectors, the equation of each flux, how the plan area is divided, how the vertical geometry and the initial state follow from a few properties, and how the impervious fraction should be estimated, and it closes with a worked example and a property reference.

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1 Which HRU to use

	Hydrologic_Response_Unit	Urban_HRU
File	hydrologic_response_unit.json	urban_hru.json
Surfaces	one, infiltrating	pervious + impervious
Internal routing reaches	none	one per surface
Soil parameters	one set	one set per layer
Groundwater area	A	A
Recharge conductivity	constant <code>recharge_K</code>	$K_{sat,3} k_r(S_{e,3})$
Groundwater storage	user S_s	unconfined, $S_s = \phi/b$
Calibration multipliers	—	<code>impervious_multiplier</code> , <code>K_sat_scale</code>
Icon		

Table 1: The two HRU composites. A is the plan area, ϕ the groundwater porosity and b the groundwater thickness.

Use the **HRU** for land without significant impervious cover — forest, fields, parks, hillslopes — and for cascades of units where the overland flow of one unit runs onto the next (Section 3). Use the **Urban HRU** wherever part of the land is impervious and drains directly to a stream or storm drain: it partitions the area, routes each surface separately, and its subsurface (Section 4) stays physically sensible when the water table lies well below the soil column.

2 Elements shared by both units

Both units build their subsurface from the same members: three **Soil** layers and a **Groundwater cell**, connected by infiltration, two percolation links and a recharge link. The geometry and the soil-water relations below apply to both; the units differ in the plan areas they give these members, the recharge link and the groundwater storage (Sections 3 and 4).

2.1 Vertical geometry

The user gives three levels; everything else is derived:

Member	Thickness	Bottom elevation
Catchment, Impervious_Catchment, reaches	—	z_s (<code>surface_elevation</code>)
Soil_1	$D/7$	$z_s - D/7$
Soil_2	$2D/7$	$z_s - 3D/7$
Soil_3	$4D/7$	$z_s - D$
Groundwater	b (<code>groundwater_thickness</code>)	$z_s - D - b$

D is `depth_to_groundwater`. The 1 : 2 : 4 split puts the thinnest layer at the surface, where moisture gradients are steepest. The groundwater cell sits directly below the soil column.

2.2 Soil water

Each soil layer stores water $V = \theta A_\ell d_\ell$ and has the van Genuchten–Mualem relations

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}, \quad \psi = -\frac{1}{\alpha} (S_e^{-1/m} - 1)^{1/n}, \quad k_r = \left[1 - (1 - S_e^{1/m})^m \right]^2, \quad m = 1 - \frac{1}{n}, \quad (1)$$

with S_e bounded to $[0.001, 0.999]$, the suction $|\psi|$ capped at 10^4 m, and the total head $h = z_{\text{bottom}} + d_\ell + \psi$ (plus a small excess pressure above saturation, governed by the specific storage). The layer's conductivity is $K_{\text{sat}} = K_{\text{sat_original}} \times K_{\text{sat_scale_factor}}$.

Infiltration

(surfacewater_to_soil_link, surface $\rightarrow$ Soil_1): $Q = \min(A_s, A_1) K_{\text{sat},1} (h_s - h_1) / (d_1/2) \cdot d / (d + 0.001)$, i.e. Darcy flow over half the top layer, switched off smoothly when no water is ponded.

Percolation

(soil_to_soil_link): $Q = \bar{A} K_{\text{eff}} (h_i - h_j) / \bar{d}$ with $K_{\text{eff}} = \min(K_{\text{sat},i}, K_{\text{sat},j}) k_r(\max(S_{e,i}, S_{e,j}))$, mean area $\bar{A}$ and mean thickness $\bar{d}$. Flow can be upward (capillary rise).

Evapotranspiration

is withdrawn from Soil_1 only, by the source connected to the Evapotranspiration property. With the Evapotranspiration_Time_Series (Soil) source the rate is $\text{corr_fact} \times ET_0 \times A_1 \theta / \theta_s$.

3 HRU

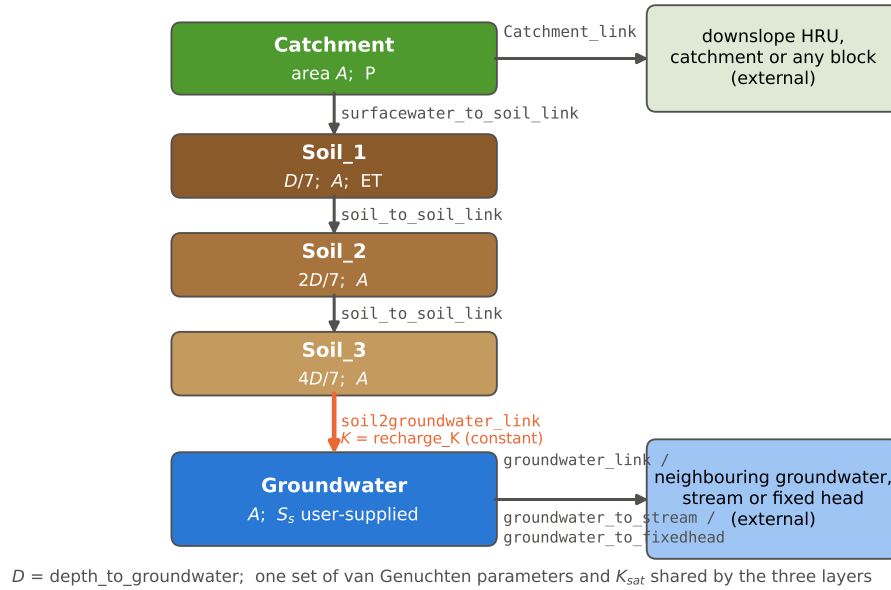


Figure 1: Hydrologic_Response_Unit: members, internal links and ports.

The HRU (hydrologic_response_unit.json, Fig. 1) has five members — a Catchment and the shared soil column and groundwater cell — and four internal links. All of them receive the whole plan area A . It has no internal reaches: the surface drains through its Catchment_link port directly to whatever block it is linked to, which makes cascades of units possible (the overland flow of an upslope unit runs onto the surface of the next). Its ports:

Port (link type)	From member	Typical target	Icon
Catchment_link	Catchment	a downslope HRU, a catchment, a channel or any block	
groundwater_link	Groundwater	the groundwater cell of a neighbouring unit	
groundwater_to_stream	Groundwater	a channel segment (baseflow)	
groundwater_to_fixedhead	Groundwater	a fixed_head block	

Recharge uses the stock `soil2groundwater_link` with the constant conductivity `recharge_K`,

$$Q_r = A K_{rech} \frac{h_3 - h_g}{L}, \quad L = \frac{1}{2} (d_3 + b), \quad (2)$$

and the groundwater cell's head is $h_g = z_g + b + (\theta_g - \phi)/S_s$ with user-supplied porosity ϕ and specific storage S_s . With the defaults ($S_s = 0.01 \text{ m}^{-1}$) the cell responds as a *confined* aquifer: a change of 0.01 in moisture content moves the head by 1 m; set $S_s = \phi/b$ for unconfined behaviour (Section 4.7). A constant recharge conductivity is adequate when the water table lies at or just below the soil column; when it lies far below, see Section 4.6. One set of soil parameters applies to all three layers. A two-unit hillslope using the HRU is in `Examples/Hydrologic_Response_Unit/HRU_hillslope.ohq`.

4 Urban HRU

4.1 Structure

Figure 2 shows the Urban HRU (`urban_hru.json`). It has eight members:

- `Catchment` — the pervious (infiltrating) surface;
- `Impervious_Catchment` — the impervious surface, with no connection to any soil;
- `Reach`, `Impervious_Reach` — two `Trapezoidal_Channel_Segments` that collect the runoff of each surface: the swales, gutters, small streams and storm pipes inside the unit;
- `Soil_1`, `Soil_2`, `Soil_3` — three van Genuchten soil layers under the pervious surface;
- `Groundwater` — a `Groundwater_cell`.

and six internal links: a `Catchment_link` from each surface to its reach, infiltration (`surfacewater_to_soil_link`) from the pervious surface into `Soil_1`, percolation (`soil_to_soil_link`) from `Soil_1` to `Soil_2` and from `Soil_2` to `Soil_3`, and recharge from `Soil_3` into the groundwater cell.

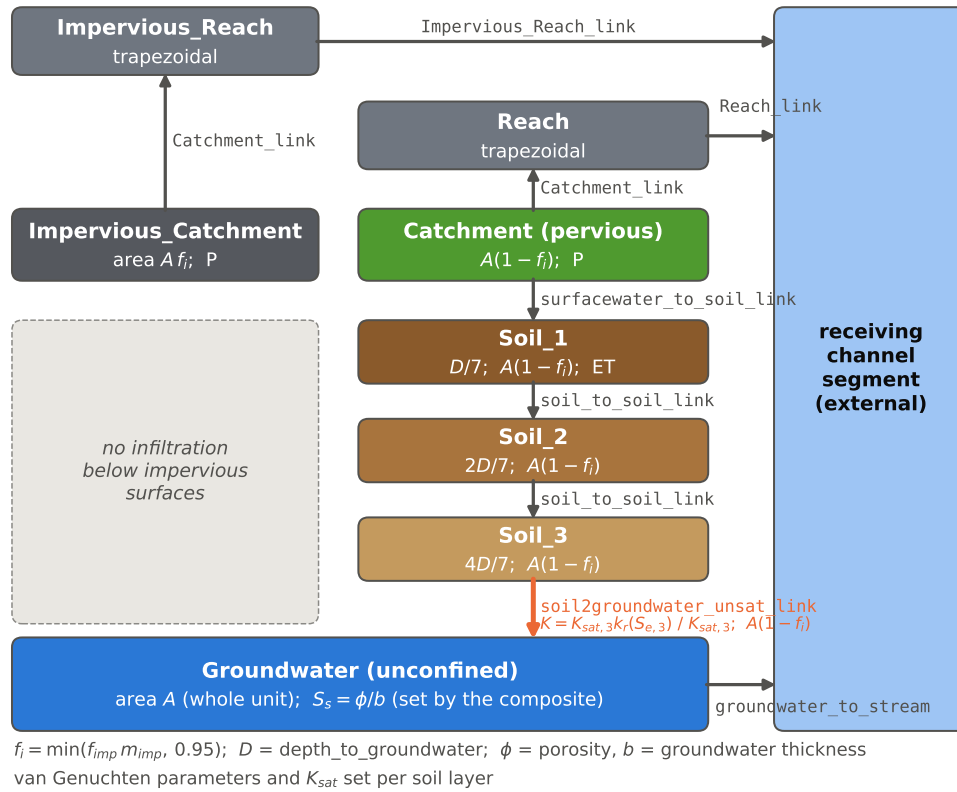


Figure 2: **Urban_HRU**: members (boxes, with thickness and plan area), internal links (arrows) and ports to the receiving channel segment. The recharge connector is highlighted.

4.2 Ports

A composite exposes the members that may be connected from outside as *ports*. The Urban HRU declares five:

Port (link type)	From member	Typical target	Icon
Reach_link	Reach	the receiving channel segment	
Impervious_Reach_link	Impervious_Reach	the receiving channel segment	
groundwater_to_stream	Groundwater	a channel segment (baseflow)	
groundwater_link	Groundwater	the groundwater cell of a neighbouring unit	
groundwater_to_fixedhead	Groundwater	a fixed_head block	

Reach_link and Impervious_Reach_link are defined in `urban_hru.json`. Their equations are those of the channel-to-channel connector (`Trapezoidal_Channel_link`); they exist as separate types only so that the two outlets of the composite can be told apart when a link is drawn from it, and they have their own icons so that HRU outlets are distinguishable from links along the stream network. When a link is drawn from the unit in the interface, the port whose type matches the chosen link is used; in a script, `create link;type=Reach_link,from=SC01,to=R01` connects the **Reach** member of SC01.

4.3 Surface runoff and routing

Rain falls on both surfaces. Water on a surface drains to its reach with the kinematic sheet-flow relation of `Catchment_link`,

$$Q_s = 86400 \frac{W}{n_s} S^{1/2} (d - d_s)_+^{5/3}, \quad (3)$$

with overland width W (`catchment_width`), slope S (`catchment_slope`), Manning coefficient n_s , ponded depth d and depression storage d_s ; $(x)_+ = \max(x, 0)$. Both surfaces share W and S and have their own n_s and d_s . The reaches drain to the receiving segment through the outlet ports with Manning's equation driven by the head difference,

$$Q_r = 86400 \frac{1}{\bar{n}} A_u R^{2/3} \left(\frac{h_r - h_c}{\bar{L}} \right)^{1/2}, \quad \bar{L} = \frac{1}{2} (L_r + L_c), \quad \bar{n} = \frac{1}{2} (n_r + n_c), \quad (4)$$

where A_u is the flow area of the end with the higher head, R the hydraulic radius from the mean perimeter, and subscripts r and c refer to the internal reach and the receiving channel. The reach bottoms are placed at the surface elevation.

4.4 The impervious fraction

The impervious surface drains directly to the internal impervious reach and from there to the channel, with no chance to infiltrate. `impervious_fraction` must therefore be the *directly connected* impervious area (DCIA) — roofs, roads and parking areas whose runoff reaches a storm drain or the stream without crossing pervious ground — and not the total impervious area (TIA) mapped from land cover. The composites take it as a plain number; how it is estimated is up to the modeller. Common choices:

- **Empirical TIA–DCIA relations.** Sutherland (2000) gives DCIA as a function of TIA (both in percent, for $\text{TIA} \geq 1\%$) for five degrees of connectivity:

Drainage	DCIA
Totally connected (curb, gutter and storm sewer; roofs tied in)	TIA
Highly connected	$0.4 \text{ TIA}^{1.2}$
Average (mostly storm-sewered, some roofs to lawns)	$0.1 \text{ TIA}^{1.5}$
Somewhat disconnected	$0.04 \text{ TIA}^{1.7}$
Highly disconnected (swales, roofs to lawns)	0.01 TIA^2

- **Storm-drain inventories.** Where inlet locations are available, impervious surfaces within a set distance of an inlet (or of a guttered street that leads to one) can be counted as connected. Public inventories miss private systems, so this tends to be a lower bound.
- **Calibration.** The estimate is scaled by `impervious_multiplier` (Section 4.9); keeping each unit's estimate and calibrating one multiplier preserves the spatial pattern and adjusts the level.

Impervious surfaces that are *not* directly connected remain part of the pervious surface (Section 4.10).

4.5 Areas

With $f_i = \min(f_{imp} m_{imp}, 0.95)$, where f_{imp} is `impervious_fraction` and m_{imp} the `impervious_multiplier`:

Member	Plan area
Impervious_Catchment	$A f_i$
Catchment, Soil_1–Soil_3, recharge interface	$A (1 - f_i)$
Groundwater	A

The aquifer continues beneath roads and roofs, so the groundwater cell receives the whole area; it is recharged only through the pervious soil column. Were it given only the pervious area, the water table would rise $1/(1 - f_i)$ times too fast per unit of recharge — twice as fast at $f_i = 0.5$ — and any change of f_i during calibration would also change the aquifer.

4.6 Recharge through the unsaturated conductivity

`soil2groundwater_unsat_link` (`groundwater.json`) connects `Soil_3` (the source, which must be a `Soil` block) to the groundwater cell:

$$Q_r = A(1 - f_i) K_{sat,3} \frac{k_r(S_{e,3}) (h_3 - h_g)_+ - (h_g - h_3)_+}{L}, \quad L = \frac{1}{2} (d_3 + b). \quad (5)$$

When the soil head is higher (drainage) the conductivity is the unsaturated conductivity of the bottom layer, with the same Mualem k_r as the percolation links; when the groundwater head is higher (upward flow) the saturated conductivity is used.

The reason is the depth of urban and upland water tables. When the water table lies several metres or more below the soil column, $h_3 - h_g$ is dominated by elevation and hardly depends on how wet the soil is. With a constant conductivity the flux is then almost independent of soil moisture: in a watershed application with the water table 6–47 m below the soil and $K_{rech} = 0.1$ m/d, the constant- K link gave a potential recharge of about 150 mm/d at $S_e = 0.5$, and at $S_e = 0.2$ the suction of the bottom layer reversed it into up to 380 mm/d of upward flow from the deep water table. Calibrating K_{rech} can only trade one error for the other. With (5) the flux falls with k_r as the bottom layer drains, and no separate recharge conductivity is needed.

4.7 Unconfined groundwater

The composite sets the cell's specific storage to $S_s = \phi/b$ (`groundwater_porosity` over `groundwater_thickness`), so that

$$h_g = z_g + b \frac{\theta_g}{\phi} : \quad (6)$$

the water table rises by $1/\phi$ per unit depth of stored water, as in an unconfined aquifer with specific yield ϕ . Because S_s is computed inside the composite, this stays true when ϕ is calibrated. To start with the water table at elevation h_0 , set `initial_groundwater_moisture_content` to $\theta_{g,0} = \phi (h_0 - z_g)/b$.

4.8 Layer-specific soil

`K_sat_i`, `alpha_vG_i`, `n_vG_i`, `theta_sat_i` and `theta_res_i` ($i = 1, 2, 3$) are set for each layer, for example from soil-survey horizons averaged over the three depth intervals. The `K_sat_i` go to each layer's `K_sat_original`, so that the layer's `K_sat_scale_factor` remains free for the multiplier below.

4.9 Calibration multipliers

Both default to 1 and leave the unit unchanged at that value.

impervious_multiplier (m_{imp})

scales the impervious fraction: $f_i = \min(f_{imp} m_{imp}, 0.95)$. It is meant for calibrating the *directly connected* impervious area watershed-wide when f_{imp} comes from an empirical estimate. $m_{imp} > 1$ means that more of the land drains straight to the channel than the estimate says (for example, more roofs and driveways tied to the storm drains); $m_{imp} < 1$ means more of the impervious surface is disconnected. Area moves between the impervious catchment and the pervious catchment with its soil column; the total stays A and the groundwater cell is unaffected. Infiltration, soil storage, Soil_1 evapotranspiration and recharge scale with the pervious area; the internal reaches and the `groundwater_to_stream` link do not change. f_{imp} itself (for example a Sutherland estimate, Section 4.4) is fixed when the model is built; only m_{imp} is calibrated.

K_sat_scale

multiplies K_{sat} of all three layers, and therefore also the recharge conductivity of (5).

Bind a model parameter to them at the composite level,

```
create parameter; type=Parameter, name=p_imp, low=0.3, high=3, value=1, prior_distribution=log-normal
set as parameter; object=SC01, quantity=impervious_multiplier, parametername=p_imp
```

and the value reaches the members through the composite's mappings. When a parameter changes, the member areas are re-evaluated and the initial soil storages $\theta_0 A_\ell d_\ell$ recomputed, so the initial moisture content stays θ_0 ; the code generator (`ohq_generate`) compiles the same mappings and initial storages as expressions of the parameter vector.

4.10 Disconnected impervious area

Only the *connected* impervious fraction is an impervious surface. Impervious surfaces that drain onto lawns — roofs with leaders to splash blocks, driveways sloping to grass — belong to the pervious catchment $A(1 - f_i)$ and are treated as soil: rain on them can infiltrate, and the Soil_1 evapotranspiration and the recharge interface act over them. Neither the concentration of their runoff onto the adjacent lawn nor the absence of transpiration from roofs is represented. If that matters, a third Catchment for the disconnected area, joined to Catchment by a Catchment_link, would represent both; it is not part of the composite.

5 Using the units

5.1 Templates

The HRU lists `main_components.json`, `unsaturated_soil.json` and `groundwater.json` in its `requires`; the Urban HRU lists the same three and `open_channel.json`, so selecting either alone in the interface is enough. The evapotranspiration source is not a member; load `soil_evapotranspiration_models.json` for the Evapotranspiration_Time_Series (Soil) source. In a script, add each template explicitly.

5.2 A worked example

Examples/Urban_HRU/Urban_HRU_example.ohq is one 50 ha unit with 25 % impervious area, draining to an 800 m trapezoidal stream that ends at a fixed head, forced by 30 days of hourly rain and daily reference ET:

```
loadtemplate; filename = main_components.json
addtemplate; filename = unsaturated_soil.json
addtemplate; filename = groundwater.json
addtemplate; filename = open_channel.json
```

```

addtemplate; filename = soil_evapotranspiration_models.json
addtemplate; filename = urban_hru.json
create source;type=Precipitation,name=Rain,timeseries=precipitation.csv
create source;type=Evapotranspiration_Time_Series (Soil),name=ET,corr_fact=0.8,ET_timeseries=
    evapotranspiration.csv
create composite;type=Urban_HRU,name=Neighbourhood,area=500000,impervious_fraction=0.25,
    surface_elevation=100,depth_to_groundwater=3.5,groundwater_thickness=12,
    Precipitation=Rain,Evapotranspiration=ET,catchment_slope=0.03,catchment_width=2500,
    K_sat_1=0.6,K_sat_2=0.4,K_sat_3=0.8,alpha_vG_1=1.5,...,
    groundwater_porosity=0.15,initial_groundwater_moisture_content=0.05,
    reach_length=400,impervious_reach_length=400,...
create block;type=Trapezoidal Channel Segment,name=Stream,base_width=3,side_slope=2,
    ManningCoeff=0.035,bottom_elevation=88,length=800,depth=0.1
create block;type=fixed_head,name=Outlet,head=87
create link;type=Reach_link,from=Neighbourhood,to=Stream,name=Pervious_outlet
create link;type=Impervious_Reach_link,from=Neighbourhood,to=Stream,name=Impervious_outlet
create link;type=groundwater_to_stream,from=Neighbourhood,to=Stream,name=Baseflow,
    area=8000,length=150,hydraulic_conductivity=5
create link;type=channel2fixed,from=Stream,to=Outlet,name=Stream_outlet

```

(Line breaks inside create commands are for display only.) Members are addressed as `Neighbourhood_Soil_3`, `Neighbourhood_recharge` and so on in outputs and scripts.

Figure 3 and Table 2 show the run. All impervious runoff reaches the stream within hours of each storm, while the pervious surface produces almost none because the rain rate rarely exceeds the infiltration capacity. About three quarters of the rain infiltrates; recharge starts at 8 mm/d as the initially wet bottom layer drains, falls with its unsaturated conductivity and settles near the rate at which water percolates through the column, and the water table rises gradually and feeds baseflow between storms.

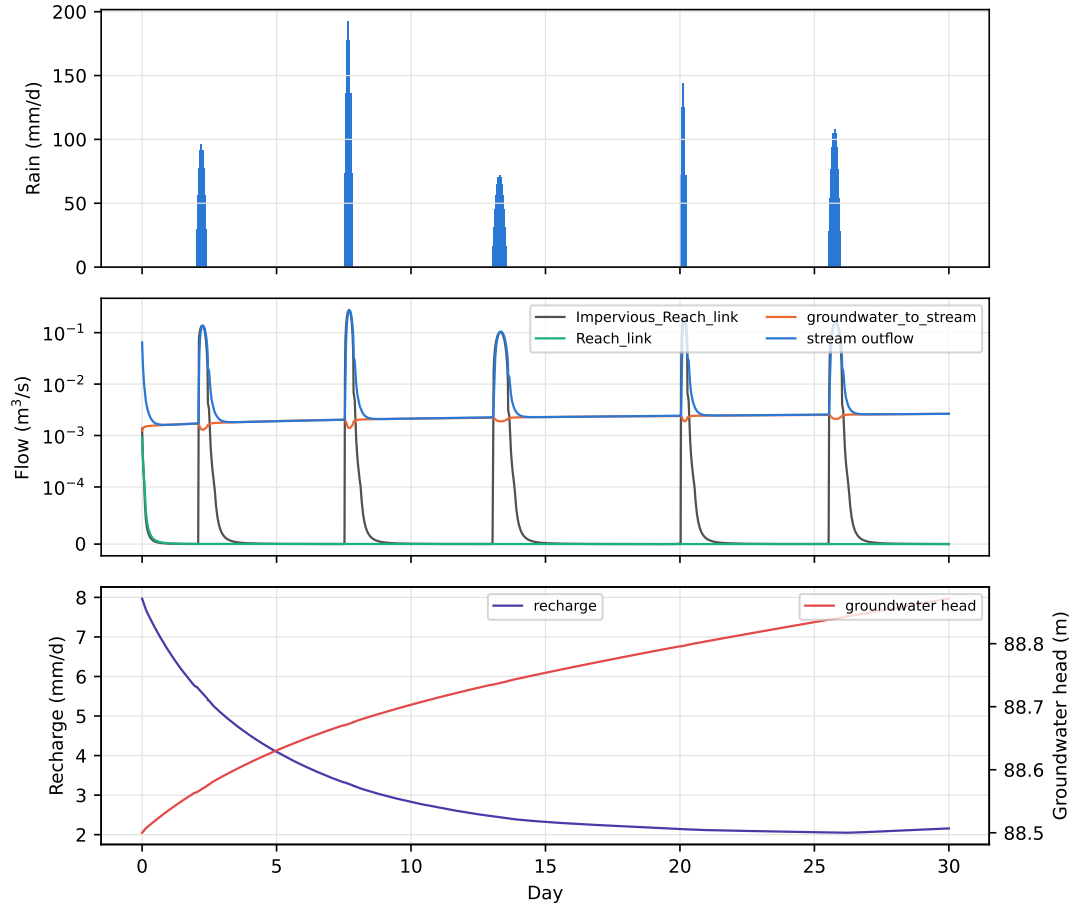


Figure 3: The worked example: rain, outflows of the three ports and of the stream, recharge and groundwater head.

Volume over 30 days	m ³
Rain	74,336
Impervious outlet (<code>Impervious_Reach_link</code>)	18,370
Pervious outlet (<code>Reach_link</code>)	4
Infiltration	55,749
Recharge	33,526
Baseflow (<code>groundwater_to_stream</code>)	5,717
Stream outflow	24,179
Groundwater head rise (m)	0.37

Table 2: Water balance of the worked example.

6 Property reference

Generated from the two resource files by `docs/make_hru_figures.py`. “In” gives the units that expose the property; “Applied to” lists the members that receive it; where the defaults differ between the units they are separated by a slash.

Property	Unit	Default	In	Applied to
<code>area</code>	m ²	1000	both	member areas
<code>recharge_K</code>	m/day	1	HRU	recharge

Property	Unit	Default	In	Applied to
surface_elevation	m	0	both	Catchment, Groundwater, Impervious_Catchment, Impervious_Reach, Reach, Soil_1, Soil_2, Soil_3
depth_to_groundwater	m	3.5	both	Soil_1, Soil_2, Soil_3
groundwater_thickness	m	5	both	Groundwater
Precipitation			both	Catchment, Impervious_Catchment
Evapotranspiration			both	Soil_1
catchment_slope		0.01	both	Catchment, Impervious_Catchment
catchment_width	m	30	both	Catchment, Impervious_Catchment
manning_coefficient		0.05	both	Catchment
depression_storage	m	0	both	Catchment
runoff_coefficient		1	both	Catchment, Impervious_Catchment
K_sat	m/day	1	HRU	Soil_1, Soil_2, Soil_3
alpha_vG	1/m	1	HRU	Soil_1, Soil_2, Soil_3
n_vG		1.41	HRU	Soil_1, Soil_2, Soil_3
theta_sat		0.4	HRU	Soil_1, Soil_2, Soil_3
theta_res		0.05	HRU	Soil_1, Soil_2, Soil_3
soil_specific_storage	1/m	0.01	both	Soil_1, Soil_2, Soil_3
initial_moisture_content		0.2	both	Soil_1, Soil_2, Soil_3
groundwater_K	m/day	1	both	Groundwater
groundwater_porosity		0.2/0.4	both	Groundwater
groundwater_specific_storage	1/m	0.01	HRU	Groundwater
initial_groundwater_moisture_content		0.1/0.25	both	Groundwater
impervious_fraction		0.2	Urban	member areas
K_sat_1	m/day	1	Urban	Soil_1
K_sat_2	m/day	1	Urban	Soil_2
K_sat_3	m/day	1	Urban	Soil_3
alpha_vG_1	1/m	1	Urban	Soil_1
alpha_vG_2	1/m	1	Urban	Soil_2
alpha_vG_3	1/m	1	Urban	Soil_3
n_vG_1		1.41	Urban	Soil_1
n_vG_2		1.41	Urban	Soil_2
n_vG_3		1.41	Urban	Soil_3
theta_sat_1		0.4	Urban	Soil_1
theta_sat_2		0.4	Urban	Soil_2
theta_sat_3		0.4	Urban	Soil_3
theta_res_1		0.05	Urban	Soil_1
theta_res_2		0.05	Urban	Soil_2
theta_res_3		0.05	Urban	Soil_3
impervious_manning_coefficient		0.015	Urban	Impervious_Catchment
impervious_depression_storage	m	0.001	Urban	Impervious_Catchment
reach_base_width	m	1	Urban	Reach
reach_side_slope		2	Urban	Reach
reach_manning_coefficient		0.035	Urban	Reach
reach_length	m	10	Urban	Reach
initial_reach_depth	m	0.01	Urban	Reach
impervious_reach_base_width	m	1	Urban	Impervious_Reach

Property	Unit	Default	In	Applied to
impervious_reach_side_slope		2	Urban	Impervious_Reach
impervious_reach_manning_coefficient		0.02	Urban	Impervious_Reach
impervious_reach_length	m	10	Urban	Impervious_Reach
initial_impervious_reach_depth	m	0.01	Urban	Impervious_Reach
impervious_multiplier		1	Urban	member areas
K_sat_scale		1	Urban	Soil_1, Soil_2, Soil_3

7 Limitations and pitfalls

- **Fixed layer split.** The three soil layers are always $D/7$, $2D/7$ and $4D/7$; D controls the storage, the recharge path length and the heads together, so calibrating it moves all three at once.
- **Deep water tables in the HRU.** When the water table is far below the soil column, the HRU's constant recharge conductivity drains the column regardless of its wetness and pulls water upward when the bottom layer dries (Section 4.6); keep `recharge_K` small or use the Urban HRU, whose recharge follows the unsaturated conductivity.
- **Confined-style defaults in the HRU.** With $S_s = 0.01 \text{ m}^{-1}$ the head responds metres to small changes of storage. For an unconfined aquifer set $S_s = \phi/b$.
- **Outlet link length.** (4) averages the length of the internal reach with that of the receiving segment, so a long receiving segment lowers the outlet slope.
- **Internal reaches are generic.** Their shape and roughness are properties of the unit, not derived from its drainage network; they mostly control the timing of the runoff peak.
- **Evapotranspiration from Soil_1 only.** Deeper layers and the groundwater cell do not transpire; the rate is scaled by θ/θ_s of the top layer.
- **Disconnected impervious area** is treated as pervious soil (Section 4.10).
- **One unit, one soil column.** Spatial variation inside a sub-catchment is represented only through its areas; use more units for more variation.

Reference

Sutherland, R. C. (2000). Methods for estimating the effective impervious area of urban watersheds. *Watershed Protection Techniques* 2(4), 282–284.